

PPS

UNIT I – Introduction to Scientific Programming & Python

Basics

- Overview of scientific computing
 - Introduction to Python programming
 - Basic syntax of Python
 - Variables and data types
 - Control structures:
 - if–else statements
 - loops
 - Functions
 - Modules
 - Introduction to MATLAB
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UNIT II – Data Structures & Algorithms

Data Structures in Python

- Lists
- Tuples
- Dictionaries

- Sets

Arrays & Matrices

- Arrays in Python
- Matrices in MATLAB

Algorithms

- Basic algorithms:
 - Searching
 - Sorting
 - Complexity analysis
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UNIT III – Numerical Methods & Linear Algebra

Numerical Methods

- Introduction to numerical methods
- Root-finding algorithms:
 - Bisection method
 - Newton–Raphson method
- Numerical integration
- Numerical differentiation

Linear Algebra

- Basics of linear algebra
 - Matrix operations
 - Solving linear systems of equations
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UNIT IV – Data Analysis & Visualization

- Importance of data analysis
 - Data manipulation using:
 - Pandas (Python)
 - MATLAB
 - Data visualization techniques
 - Introduction to plotting libraries:
 - Matplotlib
 - Seaborn (Python)
 - MATLAB plotting
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UNIT V – Emerging Technologies & Project Work

Emerging Technologies

- Overview of emerging technologies
- Machine learning

- Internet of Things (IoT)
- Big data

Machine Learning Tools

- Introduction to machine learning using:
 - Scikit-learn (Python)

Applications

- Basics of IoT and big data applications in engineering

Project & Professional Skills

- Guidelines for scientific projects
- Report writing
- Presentation skills